1 Office Salary Processing System

Scenario (Paragraph)
A private IT company processes salaries at the end of every month. Each employee's salary consists of multiple components such as Basic Pay, House Rent Allowance (HRA), and Performance Bonus. However, certain deductions like Professional Tax and Provident Fund (PF) must be subtracted before paying the final salary. The HR department wants a simple Java program where they can input these values and instantly calculate the employee's take-home salary. Your task is to help automate this calculation using variables and arithmetic operators.

Input Format

```
basic hra bonus tax pf
```

Output Format

```
Take Home Salary = value
```

Test Cases + Explanation

1. Input: 25000 8000 2000 3000 1000
   Output: 31000
   Explanation: 25000+8000+2000−3000−1000
2. 30000 10000 5000 4000 2000 → 39000
3. 18000 5000 1000 2000 500 → 21500
4. 40000 12000 8000 5000 3000 → 52000
5. 15000 4000 500 1000 500 → 18000

---

2 Mobile Data Plan Billing

Scenario
A telecom company charges customers based on mobile data usage. Every customer has a base plan cost. If they use extra data, additional charges apply per GB. A service tax is added

to the total amount. The billing team wants software that calculates the final monthly bill automatically when they enter base cost, extra GB used, rate per GB, and tax percentage.

Input Format

```
baseCost extraGB ratePerGB tax%
```

Output

```
Total Bill = value
```

Test Cases

1. 299 2 50 10 → 438.9
   Explanation: Extra=100 → Total=399 → +10%
2. 399 1 75 18 → 560.82
3. 199 0 50 5 → 208.95
4. 499 3 100 12 → 895.52
5. 249 5 40 8 → 486.72

---

3 Grocery Store Daily Sales

Scenario
A grocery shop owner wants to calculate daily earnings. Customers purchase items worth a certain amount. A discount is applied during festival season, and GST is added afterward. The owner wants a Java program to input total sales amount, discount percentage, and GST percentage to compute the final collected revenue.

Input

```
sales discount% gst%
```

Output

```
Final Revenue = value
```

Test Cases

1. 5000 10 5 → 4725
2. 3000 5 12 → 3192
3. 10000 20 18 → 9440
4. 2500 0 5 → 2625
5. 7500 15 10 → 7012.5

---

4 Student Scholarship Eligibility Amount

Scenario
A university provides scholarships based on academic performance. A student's scholarship amount is calculated from tuition fees. If the student scores above a certain level, a scholarship percentage is deducted from the fees. However, exam and library fees are still added. Build a program to compute the final payable college fee.

Input

```
tuitionFees scholarship% examFee libraryFee
```

Output

```
Final Payable Fee = value
```

Test Cases

1. 50000 20 2000 1000 → 43000
2. 60000 10 2500 1500 → 58000
3. 45000 25 1500 500 → 35500

4. 30000 0 1000 500 → 31500

5. 80000 50 3000 2000 → 45000

---

5 Restaurant Bill Splitter

Scenario
A group of friends dines at a restaurant. The total food bill is generated first. A service charge and GST are added. After that, the friends decide to split the bill equally. Write a Java program to calculate how much each person must pay.

Input

```
totalBill service% gst% numberOfPeople
```

Output

```
Amount Per Person = value
```

Test Cases

1. 2000 10 5 5 → 462
2. 1500 5 5 3 → 551.25
3. 5000 10 18 4 → 1615
4. 1200 0 5 2 → 630
5. 3000 8 12 6 → 604.8

---

6 Fuel Cost Trip Calculator

Scenario
A traveler wants to estimate fuel expenses before starting a trip. The total distance, vehicle

mileage, and fuel price per liter are known. Additionally, toll charges are added to the fuel cost. Create a Java program to calculate total travel expense.

Input

```
distance mileage fuelPrice tollCharge
```

Output

```
Total Trip Cost = value
```

Test Cases

1. 300 15 100 200 → 2200
2. 150 10 110 100 → 1750
3. 500 20 95 300 → 2675
4. 100 25 100 50 → 450
5. 600 12 105 400 → 5650

Input Format

```
daysWorked wagePerDay overtimeBonus maintenanceCharge
```

Output Format

```
Final Weekly Wage = value
```

Test Cases with Explanation

1

Input:

```
7 800 500 200
```

Output:

```
Final Weekly Wage = 5900
```

Explanation:
Base = 7×800 = 5600
Add bonus = 5600+500 = 6100
Deduct charge = 6100−200 = **5900**

2

Input:

```
6 750 0 150
```

Output:

```
Final Weekly Wage = 4350
```

Explanation:
6×750 = 4500
No bonus
4500−150 = **4350**

3

Input:

```
8 900 1000 300
```

Output:

```
Final Weekly Wage = 7900
```

Explanation:
 8×900 = 7200
 +1000 = 8200
 −300 = **7900**

4

Input:

```
5 650 0 100
```

Output:

```
Final Weekly Wage = 3150
```

Explanation:
 5×650 = 3250
 −100 = **3150**

5

Input:

```
10 1000 2000 500
```

Output:

```
Final Weekly Wage = 11500
```

Explanation:
 10×1000 = 10000+2000 = 12000−500 = **11500**

---

8 Factory Machine Production Calculator

Scenario (5+ lines)
A manufacturing factory operates heavy machines to produce steel rods daily. Each machine produces a fixed number of rods per hour. The supervisor records the total working hours and adds extra rods produced during overtime shifts. However, some rods are rejected due to quality issues and must be subtracted from total production. The factory management wants a Java program to calculate the final number of usable rods produced in a day. This helps them track real productivity accurately.

Formula

```
Final Production = (rodsPerHour × hoursWorked) + overtimeRods - rejectedRods
```

Input Format

```
rodsPerHour hoursWorked overtimeRods rejectedRods
```

Output

```
Usable Rods = value
```

Test Cases

1. 100 8 200 150 → 850
   (800 + 200 − 150)
2. 120 7 100 50 → 890
3. 90 10 300 200 → 1000
4. 150 6 0 100 → 800
5. 200 5 250 50 → 1200

9 Hospital Patient Bill Estimator

Scenario
A hospital calculates patient bills based on room charges per day and number of days admitted. Additional costs include medicine charges and lab test fees. Insurance coverage amount is deducted from the total bill. The billing department needs software to generate the final payable amount for patients quickly. This ensures transparency in hospital billing.

Formula

```
Final Bill = (roomChargePerDay × days) + medicineCost + labFees - insuranceCoverage
```

Input

```
roomCharge days medicineCost labFees insuranceCoverage
```

Output

```
Payable Amount = value
```

Test Cases

1. 2000 3 1500 800 1000 → 7300
2. 3500 2 2000 500 1500 → 8000
3. 1500 5 1000 700 500 → 8700
4. 5000 1 2500 1000 2000 → 6500
5. 2200 4 1800 900 1200 → 9300

---

10 Delivery Service Earnings Calculator

Scenario

A delivery driver earns money based on number of deliveries completed. Each delivery has a fixed payout. Incentives are added if targets are achieved. However, fuel expenses must be deducted from total earnings. The logistics company wants to compute the driver's net earnings automatically.

Formula

```
Net Earnings = (deliveries × payoutPerDelivery) + incentive - fuelCost
```

Input

```
deliveries payout incentive fuelCost
```

Output

```
Driver Earnings = value
```

Test Cases

1. 50 40 500 600 → 1900
2. 30 50 200 300 → 1400
3. 80 35 1000 900 → 2900
4. 20 60 0 200 → 1000
5. 100 30 1500 1200 → 3300

---

11 School Bus Transport Fee Calculator

Scenario
A school calculates transport fees for students based on distance traveled daily. Fee per kilometer is fixed. Maintenance charges and driver allowance are added. Government subsidy is deducted. The school office wants to automate the final transport fee calculation.

Formula

```
Transport Fee = (distance × ratePerKm) + maintenance + driverAllowance
- subsidy
```

Input

```
distance rate maintenance allowance subsidy
```

Output

```
Total Transport Fee = value
```

Test Cases

1. 10 20 200 300 100 → 600
2. 15 25 250 400 200 → 825
3. 8 30 150 200 50 → 540
4. 20 18 300 350 150 → 860
5. 12 22 180 250 80 → 614

---

12 Cinema Theatre Collection Report

Scenario
A cinema theatre tracks daily ticket sales. Revenue comes from ticket sales plus snack bar

collection. Maintenance expenses and electricity bills are deducted. Management wants to calculate net daily profit using a Java program.

Formula

```
Net Profit = ticketRevenue + snackRevenue - maintenanceCost -
electricityBill
```

Input

```
ticketRevenue snackRevenue maintenance electricity
```

Output

```
Net Profit = value
```

Test Cases

1. 50000 15000 8000 5000 → 52000
2. 30000 10000 5000 3000 → 32000
3. 75000 20000 12000 8000 → 75000
4. 20000 5000 3000 2000 → 20000
5. 90000 25000 15000 10000 → 90000

---

13 Agriculture Farm Yield Calculator

Scenario
A farmer calculates total crop yield from multiple fields. Yield per acre is multiplied by acres cultivated. Extra yield from fertilizer use is added. Damaged crops due to pests are subtracted. The farmer wants software to estimate final marketable yield.

Formula

`Final Yield = (yieldPerAcre × acres) + extraYield - damagedCrops`

Input

`yieldPerAcre acres extraYield damagedCrops`

Output

`Marketable Yield = value`

Tests

1. 500 4 200 300 → 1900
2. 650 3 150 100 → 2000
3. 400 5 300 250 → 2050
4. 700 2 0 100 → 1300
5. 550 6 400 350 → 3350

---

14 Gym Trainer Monthly Salary

Scenario
A gym trainer earns a base salary plus payment per personal training session. Performance bonus is added for high client ratings. However, equipment maintenance charges are deducted. Gym management wants an automated salary calculator.

Formula

```
Salary = baseSalary + (sessions × payPerSession) + bonus -
maintenanceFee
```

Input

```
baseSalary sessions payPerSession bonus maintenanceFee
```

Output

```
Final Salary = value
```

Tests

1. 20000 30 200 3000 1000 → 28000
2. 18000 20 250 2000 800 → 24200
3. 25000 40 150 5000 1500 → 34500
4. 15000 10 300 1000 500 → 18500
5. 22000 25 220 2500 1200 → 28800

---

15 Hotel Room Revenue Calculator

Scenario
A hotel calculates revenue from occupied rooms. Revenue depends on number of rooms booked and price per room. Extra service charges like laundry and food are added. Online booking commission is deducted. The hotel wants to compute final revenue.

Formula

```
Revenue = (roomsBooked × pricePerRoom) + serviceCharges - commission
```

Input

```
rooms price serviceCharges commission
```

Output

```
Total Revenue = value
```

Tests

1. 20 2000 15000 5000 → 50000
2. 15 1800 8000 3000 → 32000
3. 30 2500 20000 10000 → 85000
4. 10 1500 5000 2000 → 18000
5. 25 2200 12000 6000 → 61000

16 Railway Ticket Revenue System

Scenario (Paragraph)
A railway department wants to calculate daily ticket revenue for a train route. Ticket income is generated from general class passengers and AC class passengers, each having different ticket prices. Additional revenue comes from platform ticket sales. However, maintenance expenses and fuel costs must be deducted from the total collection. The railway accounts team needs a Java program to compute the final net revenue for the day. This helps them evaluate the profitability of each route efficiently.

Formula

```
Net Revenue =
(generalPassengers × generalFare)
+ (acPassengers × acFare)
+ platformCollection
- maintenanceCost
- fuelCost
```

Input

```
genPass genFare acPass acFare platform maintenance fuel
```

Output

```
Net Railway Revenue = value
```

Test Cases

1. 500 100 120 350 20000 15000 10000 → 87000
2. 300 120 80 400 15000 12000 8000 → 59800
3. 700 90 150 300 25000 20000 15000 → 103000
4. 450 110 100 320 18000 14000 9000 → 67600
5. 600 95 200 280 22000 17000 12000 → 96000

---

17 Software Company Project Cost Estimator

Scenario
A software company estimates project costs before signing clients. Cost depends on developer hours and hourly rate. UI/UX design charges and testing costs are added separately. A client advance payment is deducted from the total estimated cost. Management wants a Java system to compute the final payable project amount for budgeting clarity.

Formula

```
Project Cost =
(developerHours × hourlyRate)
+ designCost
+ testingCost
```

```
- advancePaid
```

Input

```
hours rate design testing advance
```

Output

```
Final Project Cost = value
```

Tests

1. 500 800 200000 100000 150000 → 550000
2. 300 700 120000 80000 100000 → 310000
3. 450 900 250000 150000 200000 → 605000
4. 200 600 90000 50000 40000 → 220000
5. 350 750 180000 120000 100000 → 462500

---

18 College Event Fund Management

Scenario
A college conducts an annual cultural fest. Funds are collected from student registrations and sponsorship deals. Additional income comes from stall rentals. Expenses include stage setup, celebrity payments, and marketing costs. The organizing committee wants a Java program to calculate the final remaining fund after expenses.

Formula

```
Remaining Fund =
registrationCollection
+ sponsorshipAmount
```

```
+ stallRent
- stageCost
- celebrityCost
- marketingCost
```

Input

```
registration sponsorship stall stage celebrity marketing
```

Output

```
Remaining Fund = value
```

Tests

1. 200000 500000 100000 250000 200000 50000 → 300000
2. 150000 300000 80000 200000 100000 40000 → 190000
3. 250000 600000 120000 300000 250000 70000 → 350000
4. 100000 200000 50000 120000 80000 30000 → 120000
5. 300000 700000 150000 350000 300000 90000 → 410000

---

19 Car Rental Company Profit Calculator

Scenario
A car rental agency calculates profit from rented vehicles. Revenue is generated from number of rental days multiplied by rent per day. Extra charges are collected for late returns. However, fuel refilling costs and vehicle maintenance expenses are deducted. The company needs a Java program to determine net profit per vehicle.

Formula

```
Profit =
(rentalDays × rentPerDay)
+ lateCharges
- fuelCost
- maintenanceCost
```

Input

```
days rent late fuel maintenance
```

Output

```
Net Profit = value
```

Tests

1. 10 2000 3000 4000 2000 → 17000
2. 7 1800 1500 2500 1500 → 10100
3. 15 2200 5000 6000 3000 → 29000
4. 5 2500 0 2000 1000 → 9500
5. 12 2100 2500 3500 2000 → 22200

---

20 YouTube Channel Monthly Income

Scenario

A YouTube creator earns income from ad revenue based on total views. Sponsorship income and affiliate earnings are added separately. Platform service tax and content production costs are deducted. The creator wants a Java calculator to estimate final monthly earnings.

Formula

```
Income =
adRevenue
+ sponsorshipIncome
+ affiliateIncome
- tax
- productionCost
```

Input

```
ads sponsor affiliate tax production
```

Output

```
Final YouTube Income = value
```

Tests

1. 200000 150000 50000 40000 60000 → 300000
2. 100000 80000 20000 25000 30000 → 145000
3. 300000 200000 100000 70000 90000 → 440000
4. 50000 20000 10000 8000 12000 → 60000
5. 400000 250000 150000 90000 110000 → 600000

---

21 Apartment Maintenance Collection

Scenario
An apartment association collects monthly maintenance from residents. Additional funds come from parking fees and community hall bookings. Expenses include security salaries, electricity bills, and cleaning costs. The association wants to calculate remaining balance funds using Java.

Formula

```
Balance =
maintenanceCollection
+ parkingFees
+ hallBooking
- securitySalary
- electricityBill
- cleaningCost
```

Input

```
maintenance parking hall security electricity cleaning
```

Output

```
Remaining Balance = value
```

Tests

1. 300000 50000 40000 120000 60000 30000 → 180000
2. 200000 30000 20000 90000 40000 20000 → 100000
3. 350000 60000 50000 150000 70000 40000 → 200000
4. 150000 20000 10000 60000 30000 15000 → 75000
5. 400000 80000 70000 180000 90000 50000 → 230000

---

22 Book Publishing Cost Calculator

Scenario
A publisher calculates book printing costs based on number of copies printed and cost per copy. Editing and cover design charges are added. Distributor commission and marketing expenses are deducted. The publisher needs a Java tool to compute final profit.

Formula

```
Profit =
(copies × costPerCopy)
+ editingCharge
+ designCharge
- distributorCommission
- marketingExpense
```

Input

```
copies cost editing design commission marketing
```

Output

```
Publishing Profit = value
```

Tests

1. 5000 100 50000 30000 80000 40000 → 460000
2. 3000 120 40000 20000 60000 30000 → 330000
3. 7000 90 60000 40000 100000 50000 → 580000
4. 2000 150 30000 15000 40000 20000 → 285000
5. 6000 110 55000 35000 90000 45000 → 615000

---

23 Airline Cargo Revenue System

Scenario

An airline earns cargo revenue based on total weight transported and rate per kg. Priority cargo charges and insurance fees are added. Airport handling charges and fuel surcharges are deducted. The airline finance team wants a Java calculator for final cargo revenue.

Formula

```
Revenue =
(weight × ratePerKg)
+ priorityCharges
+ insuranceFees
- handlingCharges
- fuelSurcharge
```

Input

```
weight rate priority insurance handling fuel
```

Output

```
Cargo Revenue = value
```

Tests

1. 10000 50 200000 100000 150000 120000 → 530000
2. 8000 60 150000 80000 120000 90000 → 500000
3. 12000 55 250000 120000 180000 140000 → 710000
4. 5000 70 100000 50000 80000 60000 → 360000
5. 15000 45 300000 150000 200000 180000 → 845000

---

24 Sports Tournament Prize Fund

Scenario
A sports association organizes a tournament. Funds are collected from team registrations and broadcasting rights. Sponsorship money is added. Prize distribution, stadium rent, and advertising costs are deducted. The association needs a Java system to compute remaining funds.

Formula

```
Remaining Fund =
teamRegistration
+ broadcastingRights
+ sponsorship
- prizeMoney
- stadiumRent
- advertisingCost
```

Input

```
registration broadcast sponsor prize rent advertising
```

Output

```
Remaining Tournament Fund = value
```

Tests

1. 500000 800000 400000 600000 200000 100000 → 800000
2. 300000 500000 200000 400000 150000 80000 → 370000
3. 700000 900000 500000 800000 250000 150000 → 900000
4. 200000 300000 100000 250000 100000 50000 → 200000
5. 600000 850000 450000 700000 220000 120000 → 860000

---

25 Smart City Water Supply Revenue

Scenario
A municipal corporation calculates water supply revenue for a smart city. Revenue is generated from residential and commercial water usage billed at different rates. Connection fees and late payment penalties are added. However, pipeline maintenance and water treatment costs are deducted. The corporation wants a Java program to compute final net revenue from water supply services.

Formula

```
Net Revenue =
(residentialUsage × residentialRate)
+ (commercialUsage × commercialRate)
+ connectionFees
+ penaltyCollection
- pipelineMaintenance
- treatmentCost
```

Input

```
resUsage resRate comUsage comRate connection penalty maintenance
treatment
```

Output

```
Net Water Revenue = value
```

Tests

1. 50000 5 20000 8 300000 50000 200000 150000 → 550000
2. 40000 6 15000 9 250000 40000 180000 120000 → 430000
3. 60000 5 25000 7 350000 60000 220000 180000 → 645000

4. 30000 4 10000 6 150000 20000 120000 90000 → 210000

5. 70000 6 30000 8 400000 70000 260000 200000 → 860000